M2C Edge - Link

Universal Machine to Cloud Connecting EDGE Device

Industry Partner:

M/s. ENMAZ Engineering Services Pvt. Ltd., Bengaluru

Introduction

- **M2C Edge-Link** is an advanced **Industrial IoT edge gateway** developed in collaboration with the **Central Manufacturing Technology Institute**, aimed at enabling **real-time monitoring and analytics of CNC machines** in smart manufacturing environments.
- It acts as a **bridge between shop-floor machines and server**, allowing seamless data acquisition, processing, and visualization across multiple industrial protocols such as **OPC UA, FOCAS, MODBUS, and RS485**.
- Built on a **robust Raspberry Pi CM4-based architecture**, M2C Edge-Link supports **high-frequency data collection (100ms intervals)** from CNC controllers like **SINUMERIK 840D SL**, enabling accurate tracking of machine performance.
- The system integrates **edge computing with cloud synchronization**, providing:
    i. Real-time machine status monitoring
    ii. Automated **OEE (Overall Equipment Effectiveness)** analytics
    iii. By digitizing machine data and enabling centralized access, M2C Edge-Link plays a key role in transforming traditional manufacturing setups into **smart, connected factories**.

Objective

The Edge-Link addresses the critical need for factories to digitizeM2C legacy and diverse machinery by providing a universal gateway that seamlessly connects any machine or PLC to the cloud.

Key Features

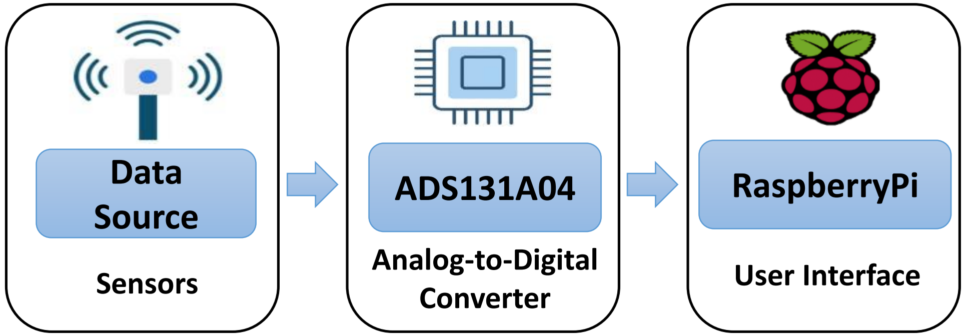
- Role-based authentication dashboard (operator/manager access levels)
- Live machine visualization with real-time status monitoring.
- Integrated email-based support ticketing system.
- Document upload functionality for job drawings and technical specifications.
- Automated OEE calculation engine with performance analytics.
- On-demand report generation for data-driven decision making.
- STM32 based 4 channel Data Acquisition for the time-domain waveform and the FFT spectrum analysis.

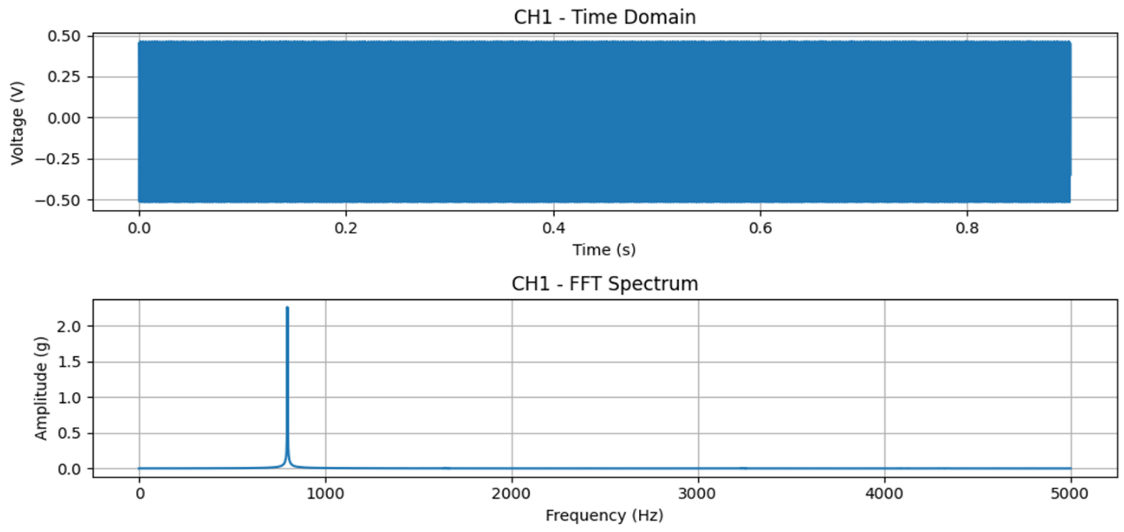
Vibration Data Acquisition

1.High-precision data acquisition: An STM32 microcontroller interfaces with a 24-bit ADS131A04 ADC to simultaneously sample four channels at 10 ksps.

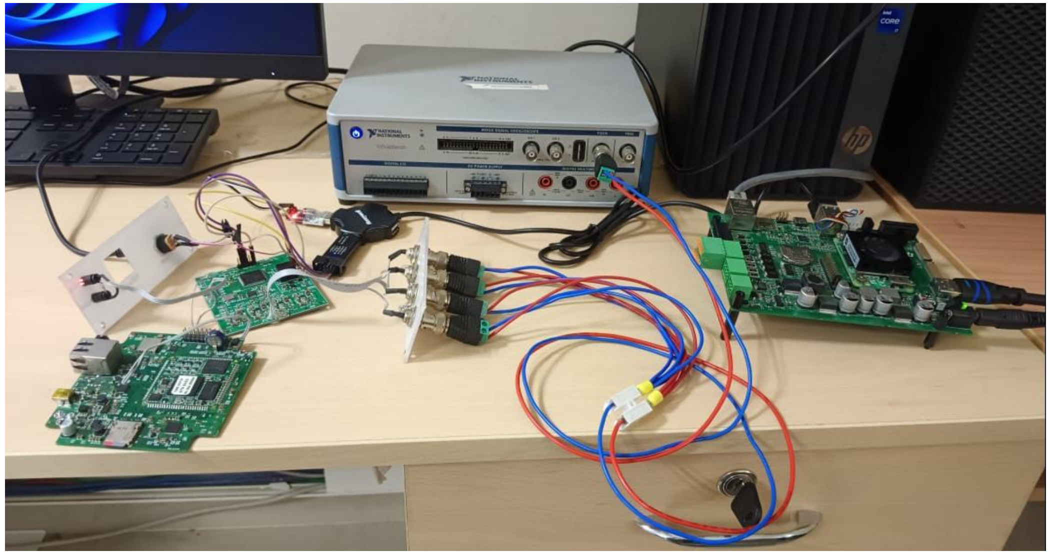
2.Edge processing: Acquired data is transmitted via UART to a Raspberry Pi CM4, where a Python script performs FFT computation.

3.Real-time visualization: The time-domain and FFT spectrum are displayed on a Node-RED dashboard for live monitoring.



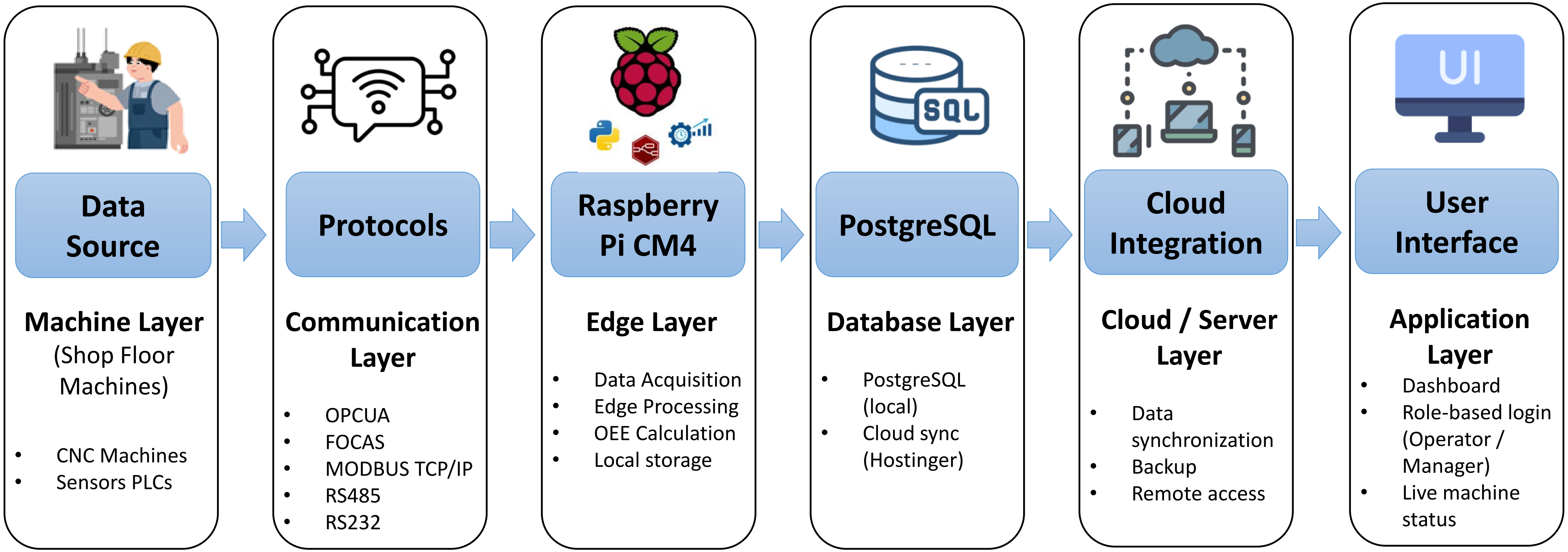


Time-domain waveform and the FFT spectrum

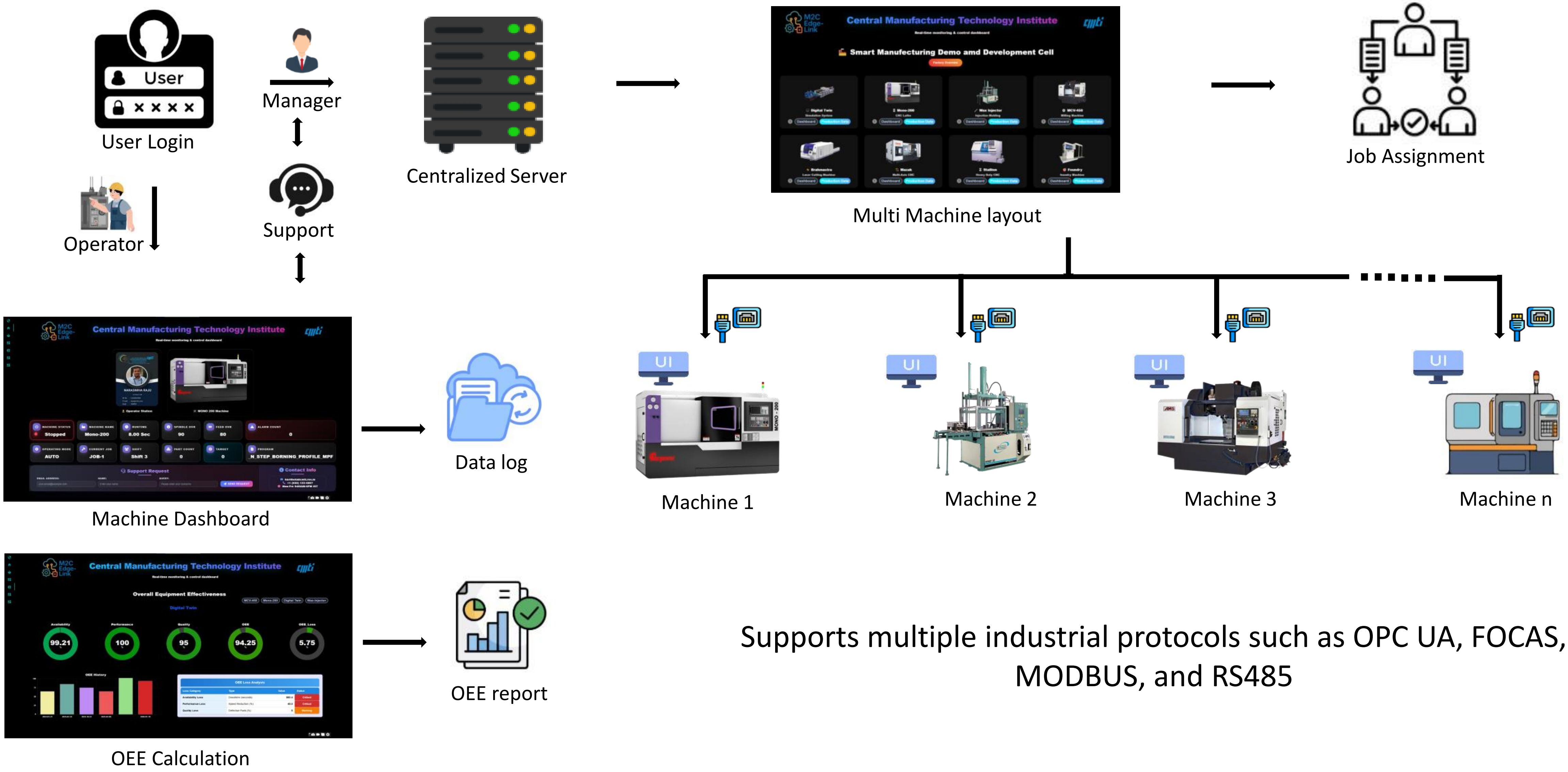


Test Setup

System Architecture



User Interface

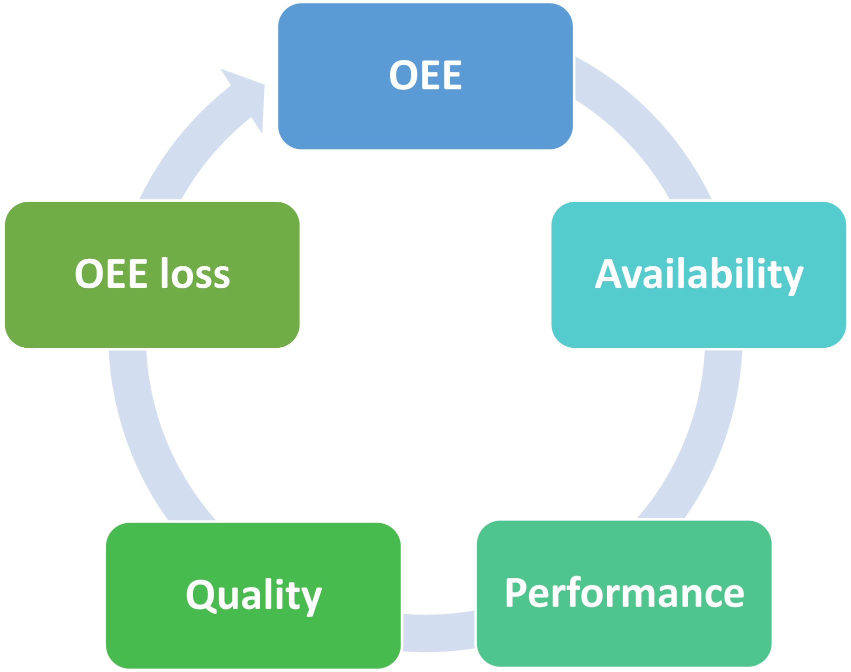


Supports multiple industrial protocols such as OPC UA, FOCAS, MODBUS, and RS485

OEE (Overall Equipment Effectiveness) Calculation

A key performance indicator that measures manufacturing productivity.

Purpose of OEE:

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OEE = (Availability × Performance × Quality) / 10000
OEE Loss = 100 - OEE
Availability = (Run Time / Production Time) × 100
Performance = ((Target Parts × Cycle Time) / Run Time) × 100
Quality = (Good Parts / Target Parts) × 100
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